

SN54HC163, SN74HC163 4-BIT SYNCHRONOUS BINARY COUNTERS

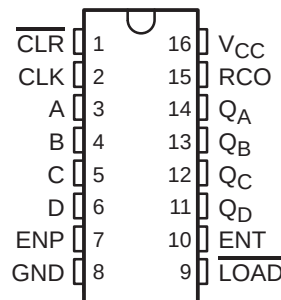
SCLS298D – JANUARY 1996 – REVISED – OCTOBER 2003

- Wide Operating Voltage Range of 2 V to 6 V
- Outputs Can Drive Up To 10 LSTTL Loads
- Low Power Consumption, 80- μ A Max I_{CC}
- Typical $t_{pd} = 14$ ns
- ± 4 -mA Output Drive at 5 V
- Low Input Current of 1 μ A Max
- Internal Look-Ahead for Fast Counting
- Carry Output for n-Bit Cascading
- Synchronous Counting
- Synchronously Programmable

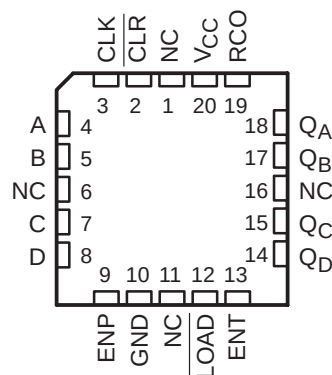
description/ordering information

These synchronous, presettable counters feature an internal carry look-ahead for application in high-speed counting designs. The 'HC163 devices are 4-bit binary counters. Synchronous operation is provided by having all flip-flops clocked simultaneously so that the outputs change coincident with each other when instructed by the count-enable (ENP, ENT) inputs and internal gating. This mode of operation eliminates the output counting spikes normally associated with synchronous (ripple-clock) counters. A buffered clock (CLK) input triggers the four flip-flops on the rising (positive-going) edge of the clock waveform.

SN54HC163 . . . J OR W PACKAGE
SN74HC163 . . . D, DB, N, NS, OR PW PACKAGE
(TOP VIEW)



SN54HC163 . . . FK PACKAGE
(TOP VIEW)



NC – No internal connection

ORDERING INFORMATION

TA	PACKAGE†		ORDERABLE PART NUMBER	TOP-SIDE MARKING
–40°C to 85°C	PDIP – N	Tube of 25	SN74HC163N	SN74HC163N
	SOIC – D	Tube of 40	SN74HC163D	HC163
		Reel of 2500	SN74HC163DR	
		Reel of 250	SN74HC163DT	
	SOP – NS	Reel of 2000	SN74HC163NSR	HC163
	SSOP – DB	Reel of 2000	SN74HC163DBR	HC163
–55°C to 125°C	TSSOP – PW	Tube of 90	SN74HC163PW	HC163
		Reel of 2000	SN74HC163PWR	
		Reel of 250	SN74HC163PWT	
	CDIP – J	Tube of 25	SNJ54HC163J	SNJ54HC163J
	CFP – W	Tube of 150	SNJ54HC163W	SNJ54HC163W
	LCCC – FK	Tube of 55	SNJ54HC163FK	SNJ54HC163FK

† Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.



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PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

 **TEXAS
INSTRUMENTS**

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On products compliant to MIL-PRF-38535, all parameters are tested unless otherwise noted. On all other products, production processing does not necessarily include testing of all parameters.

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4-BIT SYNCHRONOUS BINARY COUNTERS

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description/ordering information (continued)

These counters are fully programmable; that is, they can be preset to any number between 0 and 9 or 15. As presetting is synchronous, setting up a low level at the load input disables the counter and causes the outputs to agree with the setup data after the next clock pulse, regardless of the levels of the enable inputs.

The clear function for the 'HC163 devices is synchronous. A low level at the clear ($\overline{\text{CLR}}$) input sets all four of the flip-flop outputs low after the next low-to-high transition of CLK, regardless of the levels of the enable inputs. This synchronous clear allows the count length to be modified easily by decoding the Q outputs for the maximum count desired. The active-low output of the gate used for decoding is connected to $\overline{\text{CLR}}$ to synchronously clear the counter to 0000 (LLLL).

The carry look-ahead circuitry provides for cascading counters for n-bit synchronous applications without additional gating. ENP, ENT, and a ripple-carry output (RCO) are instrumental in accomplishing this function. Both ENP and ENT must be high to count, and ENT is fed forward to enable RCO. Enabling RCO produces a high-level pulse while the count is maximum (9 or 15 with Q_A high). This high-level overflow ripple-carry pulse can be used to enable successive cascaded stages. Transitions at ENP or ENT are allowed, regardless of the level of CLK.

These counters feature a fully independent clock circuit. Changes at control inputs (ENP, ENT, or $\overline{\text{LOAD}}$) that modify the operating mode have no effect on the contents of the counter until clocking occurs. The function of the counter (whether enabled, disabled, loading, or counting) is dictated solely by the conditions meeting the stable setup and hold times.

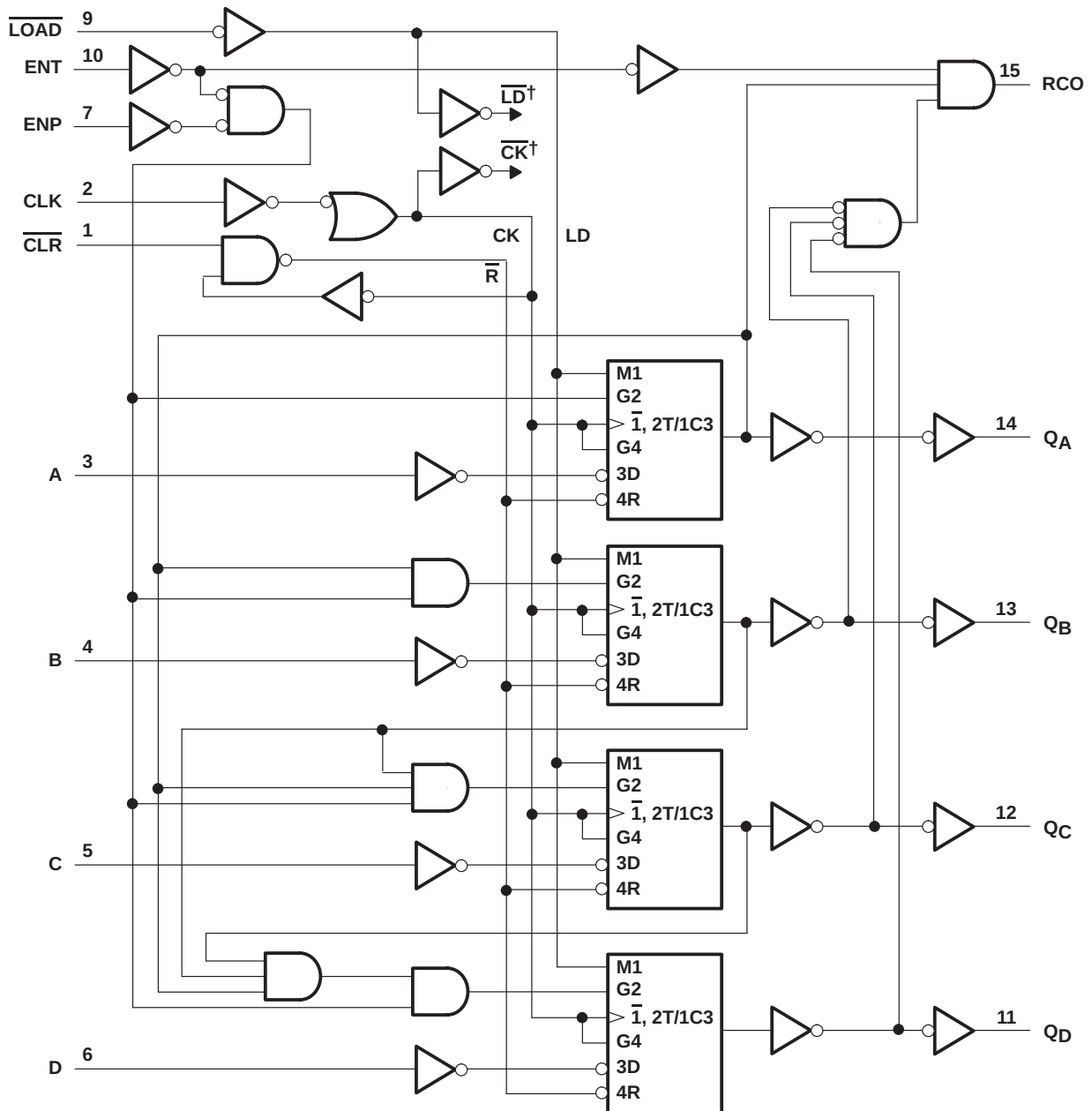


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logic diagram (positive logic)



† For simplicity, routing of complementary signals $\overline{LD}$ and $\overline{CK}$ is not shown on this overall logic diagram. The uses of these signals are shown on the logic diagram of the D/T flip-flops.

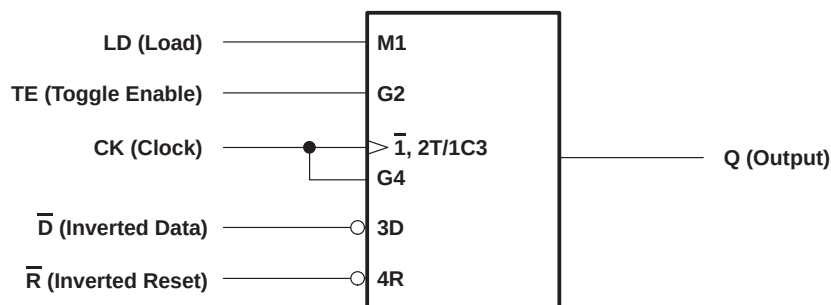
Pin numbers shown are for the D, DB, J, N, NS, PW, and W packages.

SN54HC163, SN74HC163

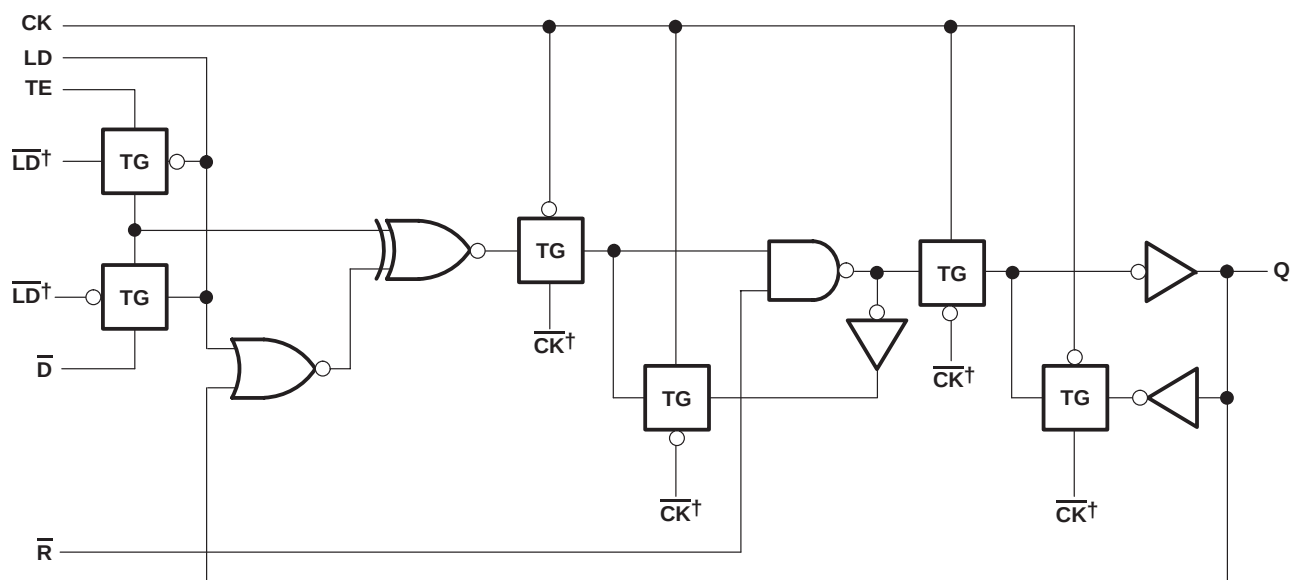
4-BIT SYNCHRONOUS BINARY COUNTERS

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logic symbol, each D/T flip-flop



logic diagram, each D/T flip-flop (positive logic)

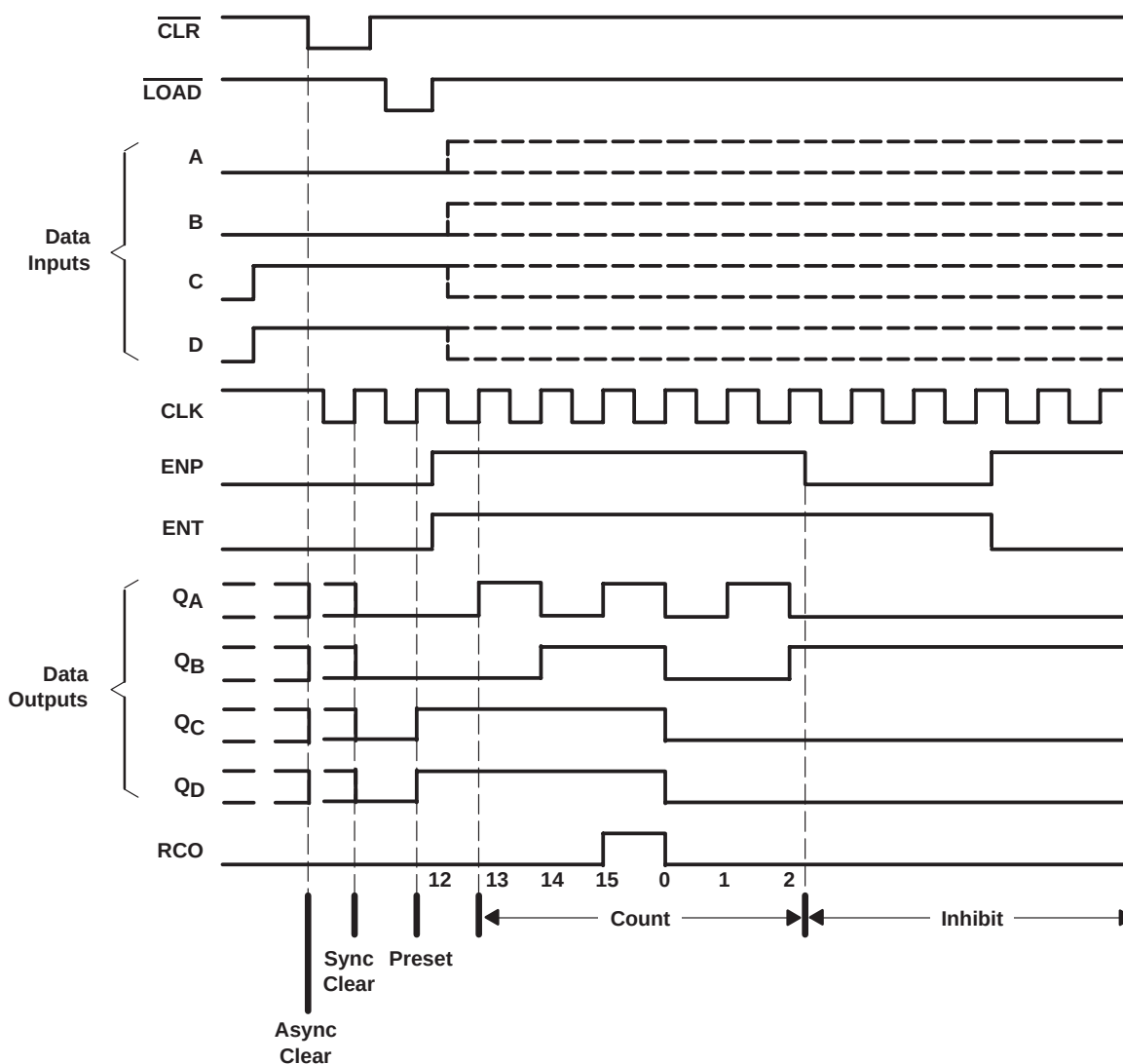


† The origins of $\overline{LD}$ and $\overline{CK}$ are shown in the logic diagram of the overall device.

typical clear, preset, count, and inhibit sequence

The following sequence is illustrated below:

1. Clear outputs to zero (synchronous)
2. Preset to binary 12
3. Count to 13, 14, 15, 0, 1, and 2
4. Inhibit



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absolute maximum ratings over operating free-air temperature range (unless otherwise noted)[†]

Supply voltage range, V_{CC}	–0.5 V to 7 V
Input clamp current, I_{IK} ($V_I < 0$ or $V_I > V_{CC}$) (see Note 1)	±20 mA
Output clamp current, I_{OK} ($V_O < 0$ or $V_O > V_{CC}$) (see Note 1)	±20 mA
Continuous output current, I_O ($V_O = 0$ to V_{CC})	±25 mA
Continuous current through V_{CC} or GND	±50 mA
Package thermal impedance, θ_{JA} (see Note 2): D package	73°C/W
DB package	82°C/W
N package	67°C/W
NS package	64°C/W
PW package	108°C/W
Storage temperature range, T_{stg}	–65°C to 150°C

[†] Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

NOTES: 1. The input and output voltage ratings may be exceeded if the input and output current ratings are observed.
2. The package thermal impedance is calculated in accordance with JESD 51-7.

recommended operating conditions (see Note 3)

			SN54HC163			SN74HC163			UNIT
			MIN	NOM	MAX	MIN	NOM	MAX	
V _{CC}	Supply voltage		2	5	6	2	5	6	V
V _{IH}	High-level input voltage	V _{CC} = 2 V	1.5			1.5			V
		V _{CC} = 4.5 V	3.15			3.15			
		V _{CC} = 6 V	4.2			4.2			
V _{IL}	Low-level input voltage	V _{CC} = 2 V	0.5			0.5			V
		V _{CC} = 4.5 V	1.35			1.35			
		V _{CC} = 6 V	1.8			1.8			
V _I	Input voltage		0	V _{CC}		0	V _{CC}		V
V _O	Output voltage		0	V _{CC}		0	V _{CC}		V
Δt/Δv [‡]	Input transition rise/fall time	V _{CC} = 2 V	1000			1000			ns
		V _{CC} = 4.5 V	500			500			
		V _{CC} = 6 V	400			400			
T _A	Operating free-air temperature		−55	125		−40	85		°C

NOTE 3: All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.

[‡] If this device is used in the threshold region (from $V_{ILmax} = 0.5$ V to $V_{IHmin} = 1.5$ V), there is a potential to go into the wrong state from induced grounding, causing double clocking. Operating with the inputs at $t_t = 1000$ ns and $V_{CC} = 2$ V does not damage the device; however, functionally, the CLK inputs are not ensured while in the shift, count, or toggle operating modes.



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electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS		V _{CC}	T _A = 25°C			SN54HC163		SN74HC163		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
V _{OH}	V _I = V _{IH} or V _{IL}	I _{OH} = -20 µA	2 V	1.9	1.998		1.9		1.9		V
			4.5 V	4.4	4.499		4.4		4.4		
			6 V	5.9	5.999		5.9		5.9		
		I _{OH} = -4 mA	4.5 V	3.98	4.3		3.7		3.84		
		I _{OH} = -5.2 mA	6 V	5.48	5.8		5.2		5.34		
V _{OL}	V _I = V _{IH} or V _{IL}	I _{OL} = 20 µA	2 V		0.002	0.1		0.1		0.1	V
			4.5 V		0.001	0.1		0.1		0.1	
			6 V		0.001	0.1		0.1		0.1	
		I _{OL} = 4 mA	4.5 V		0.17	0.26		0.4		0.33	
		I _{OL} = 5.2 mA	6 V		0.15	0.26		0.4		0.33	
I _I	V _I = V _{CC} or 0		6 V		±0.1	±100		±1000		±1000	nA
I _{CC}	V _I = V _{CC} or 0, I _O = 0		6 V			8		160		80	µA
C _i			2 V to 6 V		3	10		10		10	pF

timing requirements over recommended operating free-air temperature range (unless otherwise noted)

			V _{CC}	T _A = 25°C		SN54HC163		SN74HC163		UNIT
				MIN	MAX	MIN	MAX	MIN	MAX	
f _{clock}	Clock frequency		2 V	6		4.2		5		MHz
			4.5 V	31		21		25		
			6 V	36		25		29		
t _w	Pulse duration	CLK high or low	2 V	80		120		100		ns
			4.5 V	16		24		20		
			6 V	14		20		17		
t _{su}	A, B, C, or D		2 V	150		225		190		ns
			4.5 V	30		45		38		
			6 V	26		38		32		
	$\overline{\text{LOAD}}$ low		2 V	135		205		170		
			4.5 V	27		41		34		
			6 V	23		35		29		
	ENP, ENT		2 V	170		255		215		
			4.5 V	34		51		43		
			6 V	29		43		37		
	$\overline{\text{CLR}}$ low		2 V	160		240		200		
			4.5 V	32		48		40		
			6 V	27		41		34		
	$\overline{\text{CLR}}$ inactive		2 V	160		240		200		
			4.5 V	32		48		40		
			6 V	27		41		34		
t _h	Hold time, all synchronous inputs after CLK↑		2 V	0		0		0		ns
			4.5 V	0		0		0		
			6 V	0		0		0		



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4-BIT SYNCHRONOUS BINARY COUNTERS

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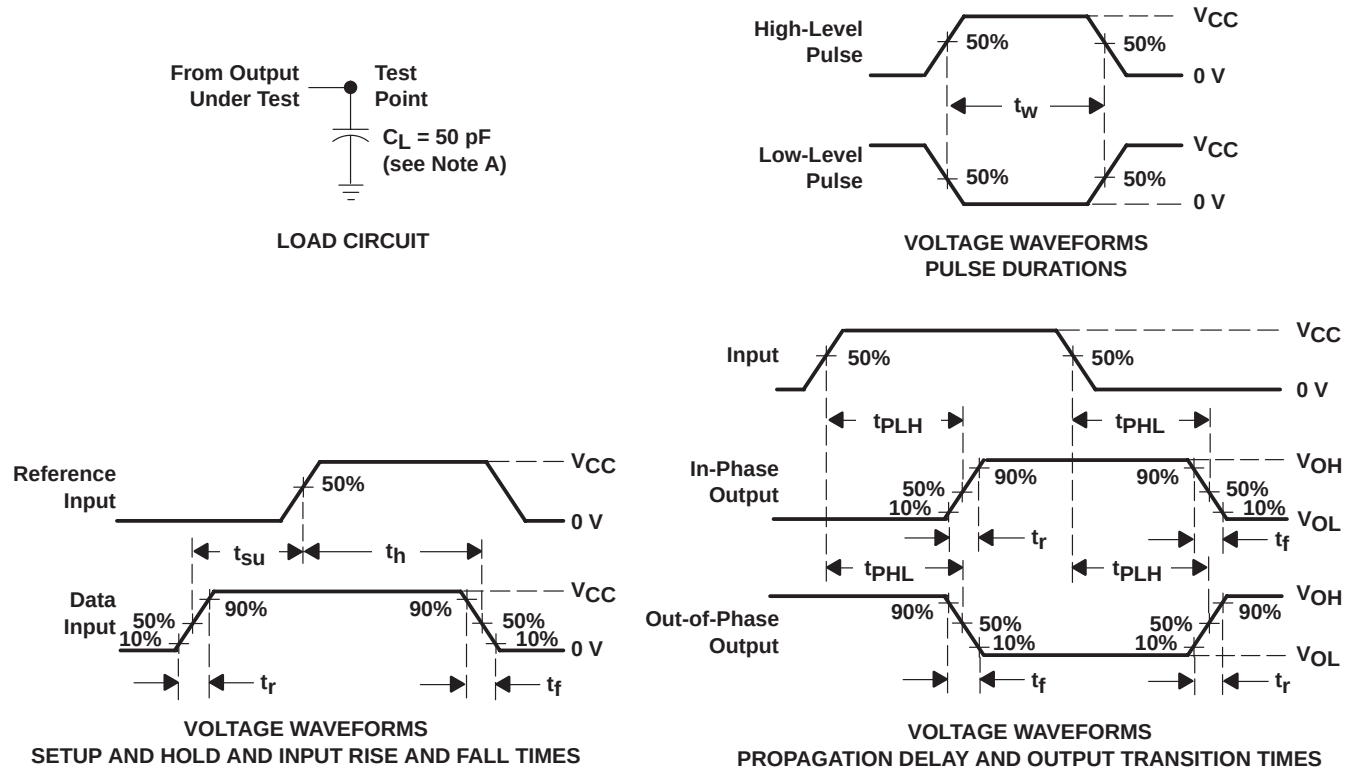
switching characteristics over recommended operating free-air temperature range, $C_L = 50$ pF (unless otherwise noted) (see Figure 1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CC}	$T_A = 25^\circ\text{C}$			SN54HC163		SN74HC163		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
$f_{\max}$			2 V	6	14		4.2		5		MHz
			4.5 V	31	40		21		25		
			6 V	36	44		25		29		
t_{pd}	CLK	RCO	2 V		83	215		325		270	ns
			4.5 V		24	43		65		54	
			6 V		20	37		55		46	
		Any Q	2 V		80	205		310		255	
			4.5 V		25	41		62		51	
			6 V		21	35		53		43	
	ENT	RCO	2 V		62	195		295		245	
			4.5 V		17	39		59		49	
			6 V		14	33		50		42	
t_t		Any	2 V		38	75		110		95	ns
			4.5 V		8	15		22		19	
			6 V		6	13		19		16	

operating characteristics, $T_A = 25^\circ\text{C}$

PARAMETER		TEST CONDITIONS	TYP	UNIT
C_{pd}	Power dissipation capacitance	No load	60	pF

PARAMETER MEASUREMENT INFORMATION



- NOTES:
- A. C_L includes probe and test-fixture capacitance.
 - B. Phase relationships between waveforms were chosen arbitrarily. All input pulses are supplied by generators having the following characteristics: $PRR \leq 1 \text{ MHz}$, $Z_O = 50 \Omega$, $t_r = 6 \text{ ns}$, $t_f = 6 \text{ ns}$.
 - C. For clock inputs, f_{max} is measured when the input duty cycle is 50%.
 - D. The outputs are measured one at a time with one input transition per measurement.
 - E. t_{PLH} and t_{PHL} are the same as t_{pd} .

Figure 1. Load Circuit and Voltage Waveforms

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4-BIT SYNCHRONOUS BINARY COUNTERS

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APPLICATION INFORMATION

n-bit synchronous counters

This application demonstrates how the look-ahead carry circuit can be used to implement a high-speed n-bit counter. The 'HC163 devices count in binary. Virtually any count mode (modulo-N, N_1 -to- N_2 , N_1 -to-maximum) can be used with this fast look-ahead circuit.

The application circuit shown in Figure 2 is not valid for clock frequencies above 18 MHz (at 25°C and 4.5-V V_{CC}). The reason for this is that there is a glitch that is produced on the second stage's RCO and every succeeding stage's RCO. This glitch is common to all HC vendors that Texas Instruments has evaluated, in addition to the bipolar equivalents (LS, ALS, AS).



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APPLICATION INFORMATION

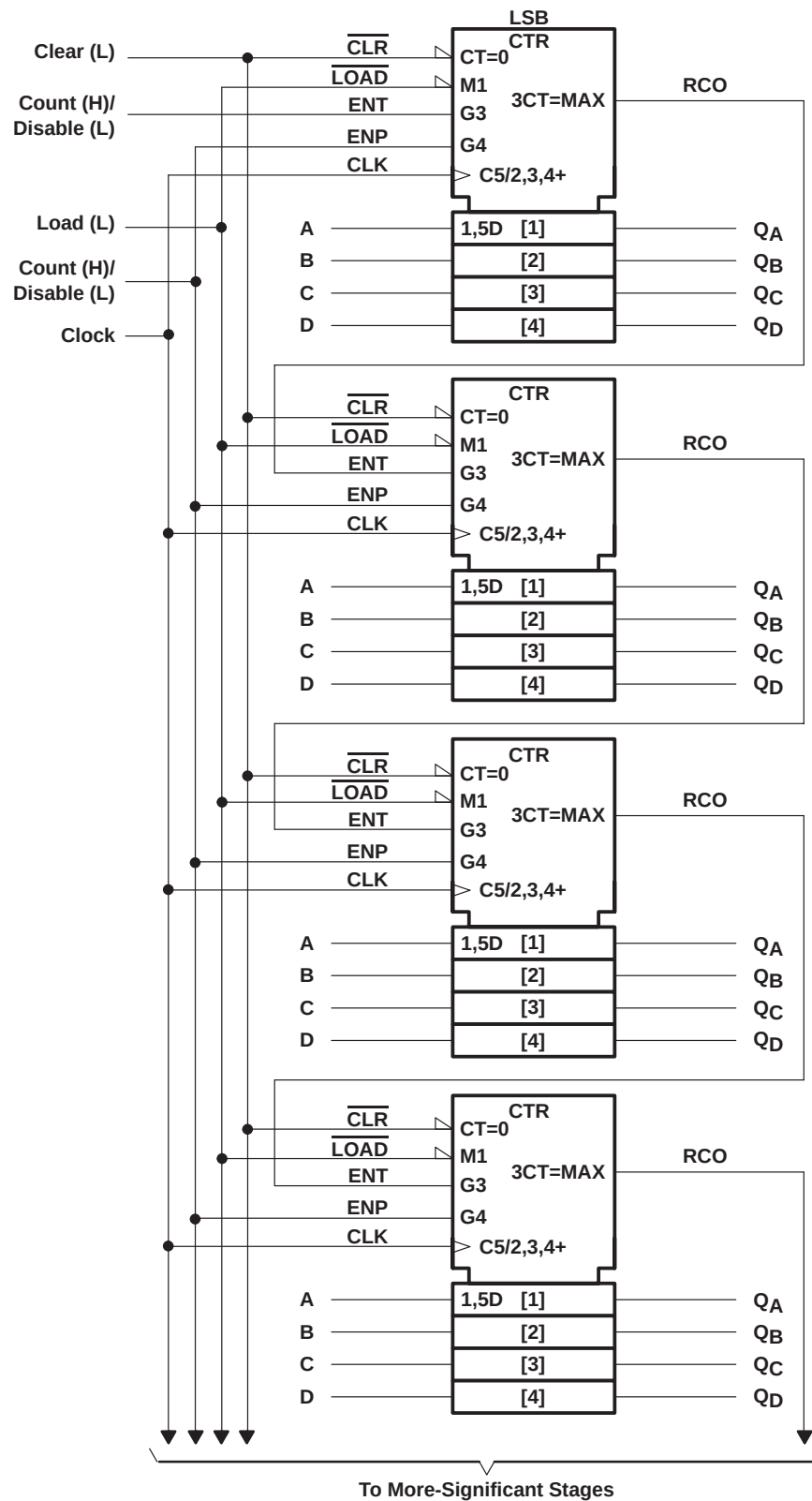


Figure 2

SN54HC163, SN74HC163

4-BIT SYNCHRONOUS BINARY COUNTERS

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APPLICATION INFORMATION

n-bit synchronous counters (continued)

The glitch on RCO is caused because the propagation delay of the rising edge of Q_A of the second stage is shorter than the propagation delay of the falling edge of ENT. RCO is the product of ENT, Q_A , Q_B , Q_C , and Q_D ($ENT \times Q_A \times Q_B \times Q_C \times Q_D$). The resulting glitch is about 7–12 ns in duration. Figure 3 shows the condition in which the glitch occurs. For simplicity, only two stages are being considered, but the results can be applied to other stages. Q_B , Q_C , and Q_D of the first and second stage are at logic one, and Q_A of both stages are at logic zero (1110 1110) after the first clock pulse. On the rising edge of the second clock pulse, Q_A and RCO of the first stage go high. On the rising edge of the third clock pulse, Q_A and RCO of the first stage return to a low level, and Q_A of the second stage goes to a high level. At this time, the glitch on RCO of the second stage appears because of the race condition inside the chip.

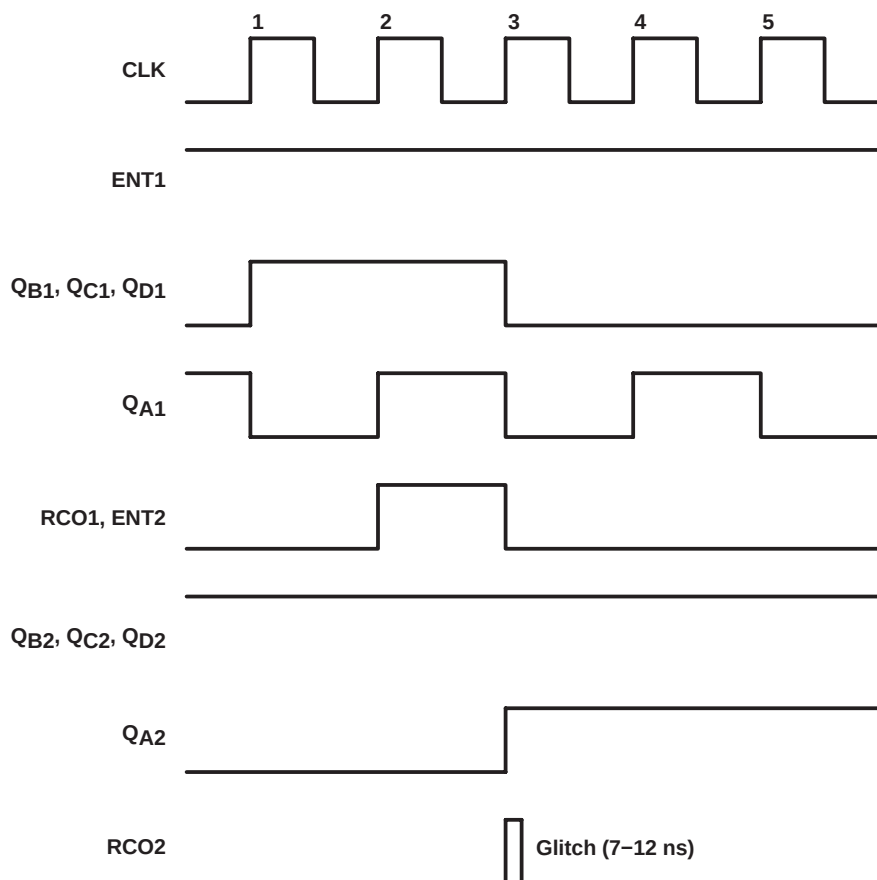


Figure 3

The glitch causes a problem in the next stage (stage three) if the glitch is still present when the next rising clock edge appears (clock pulse 4). To ensure that this does not happen, the clock frequency must be less than the inverse of the sum of the clock-to-RCO propagation delay and the glitch duration (t_g). In other words, $f_{max} = 1/(t_{pd} \text{ CLK-to-RCO} + t_g)$. For example, at 25°C at 4.5-V V_{CC} , the clock-to-RCO propagation delay is 43 ns and the maximum duration of the glitch is 12 ns. Therefore, the maximum clock frequency that the cascaded counters can use is 18 MHz. The following tables contain the f_{clock} , t_w , and f_{max} specifications for applications that use more than two 'HC163 devices cascaded together.

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APPLICATION INFORMATION

n-bit synchronous counters (continued)

timing requirements over recommended operating free-air temperature range (unless otherwise noted)

		V _{CC}	T _A = 25°C		SN54HC163		SN74HC163		UNIT
			MIN	MAX	MIN	MAX	MIN	MAX	
f _{clock}	Clock frequency	2 V	3.6		2.5		2.9		MHz
		4.5 V	18		12		14		
		6 V	21		14		17		
t _w	Pulse duration, CLK high or low	2 V	140		200		170		ns
		4.5 V	28		40		36		
		6 V	24		36		30		

switching characteristics over recommended operating free-air temperature range, $C_L = 50$ pF (unless otherwise noted) (see Note 4)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V _{CC}	T _A = 25°C		SN54HC163		SN74HC163		UNIT
				MIN	MAX	MIN	MAX	MIN	MAX	
f _{max}			2 V	3.6		2.5		2.9		MHz
			4.5 V	18		12		14		
			6 V	21		14		17		

NOTE 4: These limits apply only to applications that use more than two 'HC163 devices cascaded together.

If the 'HC163 devices are used as a single unit, or only two cascaded together, then the maximum clock frequency that the devices can use is not limited because of the glitch. In these situations, the devices can be operated at the maximum specifications.

A glitch can appear on RCO of a single 'HC163 device, depending on the relationship of ENT to CLK. Any application that uses RCO to drive any input, except an ENT of another cascaded 'HC163 device, must take this into consideration.

PACKAGING INFORMATION

Orderable Device	Status ⁽¹⁾	Package Type	Package Drawing	Pins	Package Qty	Eco Plan ⁽²⁾	Lead/Ball Finish	MSL Peak Temp ⁽³⁾
86076012A	ACTIVE	LCCC	FK	20	1	None	Call TI	Level-NC-NC-NC
8607601EA	ACTIVE	CDIP	J	16	1	None	Call TI	Level-NC-NC-NC
8607601FA	ACTIVE	CFP	W	16	1	None	Call TI	Level-NC-NC-NC
JM38510/66304BEA	ACTIVE	CDIP	J	16	1	None	Call TI	Level-NC-NC-NC
SN54HC163J	ACTIVE	CDIP	J	16	1	None	Call TI	Level-NC-NC-NC
SN74HC163D	ACTIVE	SOIC	D	16	40	Pb-Free (RoHS)	CU NIPDAU	Level-2-260C-1 YEAR/ Level-1-235C-UNLIM
SN74HC163DBR	ACTIVE	SSOP	DB	16	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-260C-1 YEAR/ Level-1-235C-UNLIM
SN74HC163DR	ACTIVE	SOIC	D	16	2500	Pb-Free (RoHS)	CU NIPDAU	Level-2-260C-1 YEAR/ Level-1-235C-UNLIM
SN74HC163DT	ACTIVE	SOIC	D	16	250	Pb-Free (RoHS)	CU NIPDAU	Level-2-260C-1 YEAR/ Level-1-235C-UNLIM
SN74HC163N	ACTIVE	PDIP	N	16	25	Pb-Free (RoHS)	CU NIPDAU	Level-NC-NC-NC
SN74HC163NSR	ACTIVE	SO	NS	16	2000	Pb-Free (RoHS)	CU NIPDAU	Level-2-260C-1 YEAR/ Level-1-235C-UNLIM
SN74HC163PW	ACTIVE	TSSOP	PW	16	90	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
SN74HC163PWR	ACTIVE	TSSOP	PW	16	2000	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
SN74HC163PWT	ACTIVE	TSSOP	PW	16	250	Pb-Free (RoHS)	CU NIPDAU	Level-1-250C-UNLIM
SNJ54HC163FK	ACTIVE	LCCC	FK	20	1	None	Call TI	Level-NC-NC-NC
SNJ54HC163J	ACTIVE	CDIP	J	16	1	None	Call TI	Level-NC-NC-NC
SNJ54HC163W	ACTIVE	CFP	W	16	1	None	Call TI	Level-NC-NC-NC

⁽¹⁾ The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

⁽²⁾ Eco Plan - May not be currently available - please check <http://www.ti.com/productcontent> for the latest availability information and additional product content details.

None: Not yet available Lead (Pb-Free).

Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.

Green (RoHS & no Sb/Br): TI defines "Green" to mean "Pb-Free" and in addition, uses package materials that do not contain halogens, including bromine (Br) or antimony (Sb) above 0.1% of total product weight.

⁽³⁾ MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

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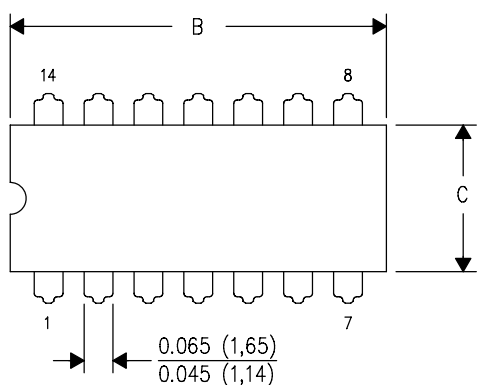
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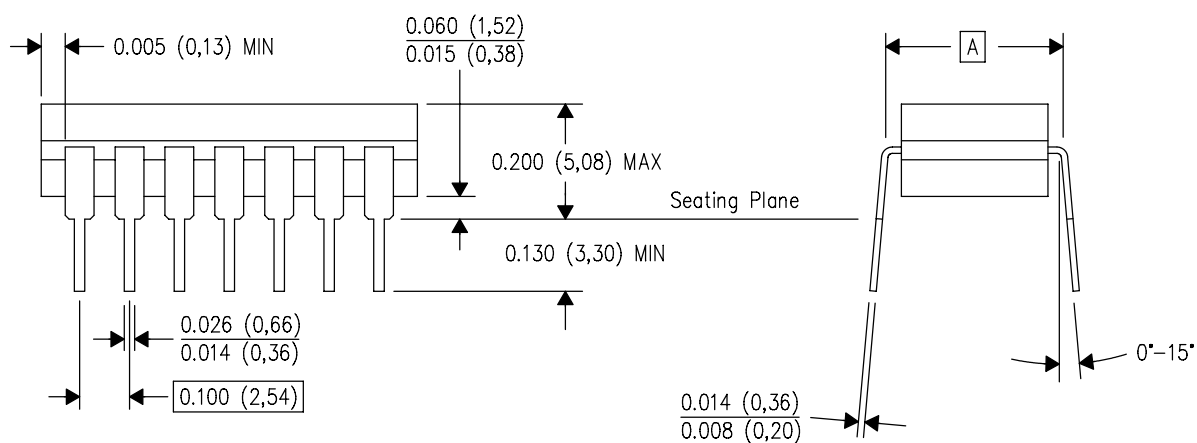
J (R-GDIP-T**)

14 LEADS SHOWN

CERAMIC DUAL IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC
B MAX	0.785 (19,94)	.840 (21,34)	0.960 (24,38)	1.060 (26,92)
B MIN	—	—	—	—
C MAX	0.300 (7,62)	0.300 (7,62)	0.310 (7,87)	0.300 (7,62)
C MIN	0.245 (6,22)	0.245 (6,22)	0.220 (5,59)	0.245 (6,22)

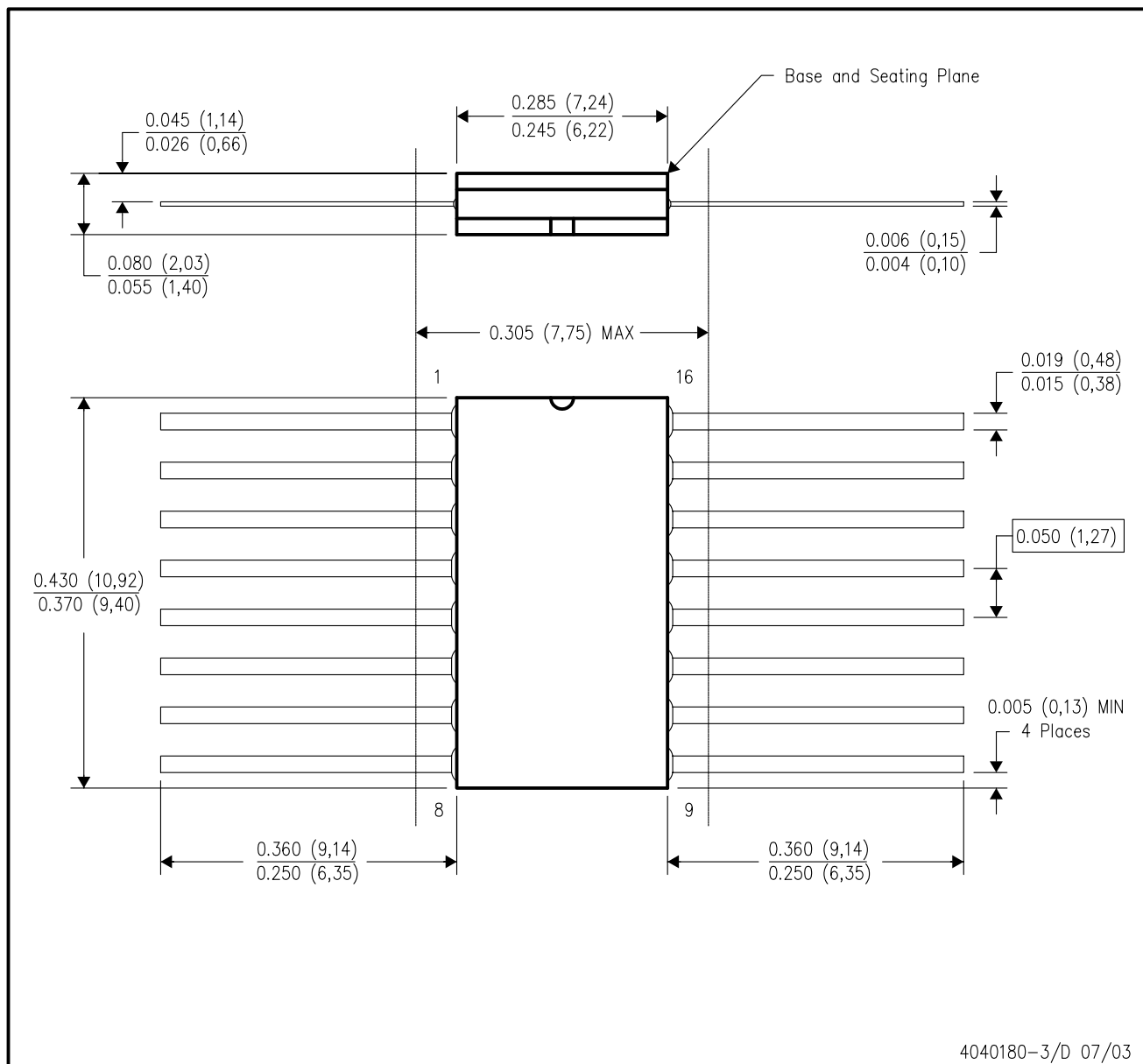


4040083/F 03/03

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. This package is hermetically sealed with a ceramic lid using glass frit.
 - D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
 - E. Falls within MIL STD 1835 GDIP1-T14, GDIP1-T16, GDIP1-T18 and GDIP1-T20.

W (R-GDFP-F16)

CERAMIC DUAL FLATPACK

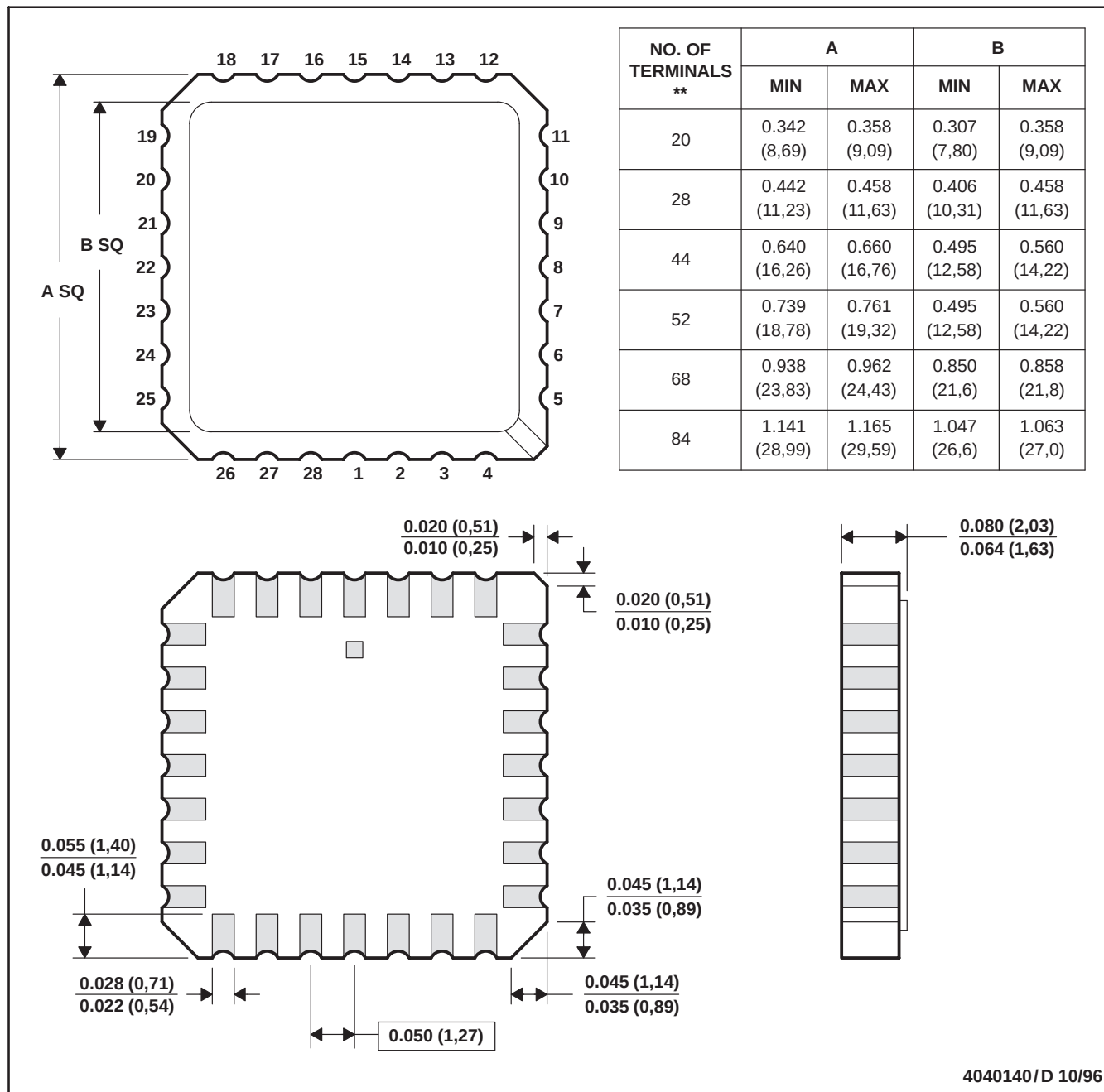


- NOTES:
- All linear dimensions are in inches (millimeters).
 - This drawing is subject to change without notice.
 - This package can be hermetically sealed with a ceramic lid using glass frit.
 - Index point is provided on cap for terminal identification only.
 - Falls within MIL STD 1835 GDFP1-F16 and JEDEC MO-092AC

FK (S-CQCC-N**)

LEADLESS CERAMIC CHIP CARRIER

28 TERMINAL SHOWN

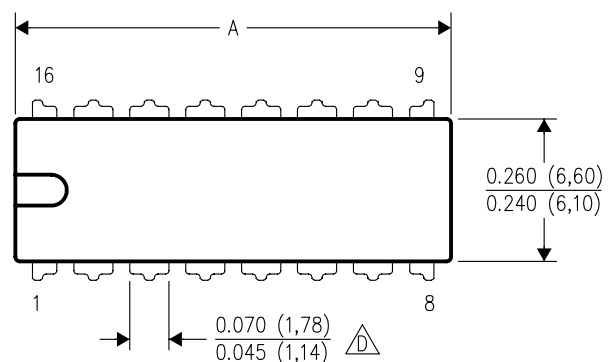


- NOTES:
- All linear dimensions are in inches (millimeters).
 - This drawing is subject to change without notice.
 - This package can be hermetically sealed with a metal lid.
 - The terminals are gold plated.
 - Falls within JEDEC MS-004

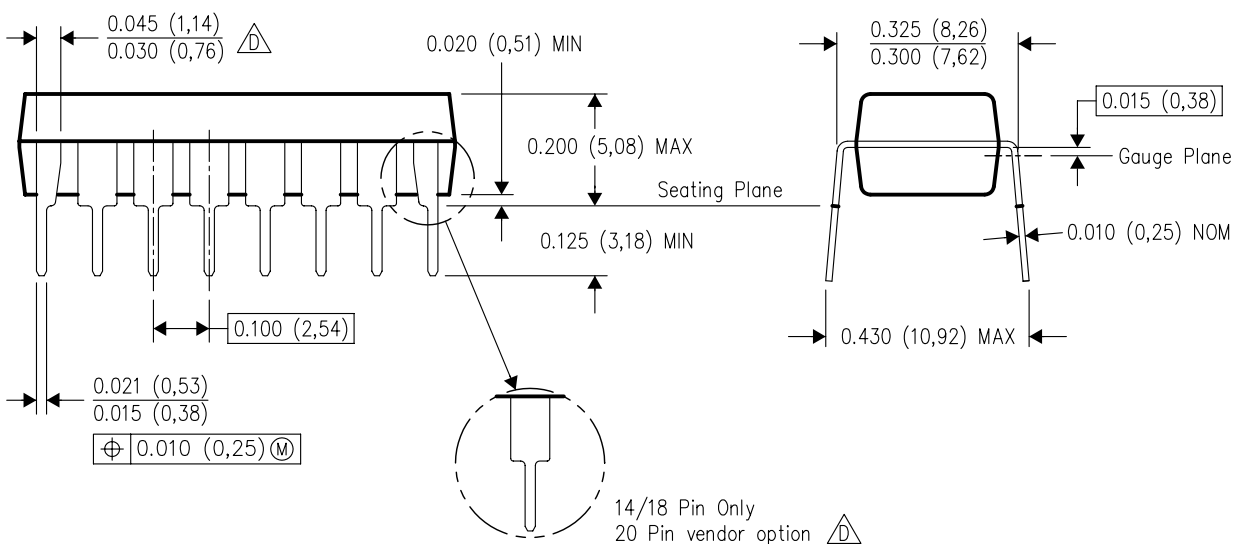
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD

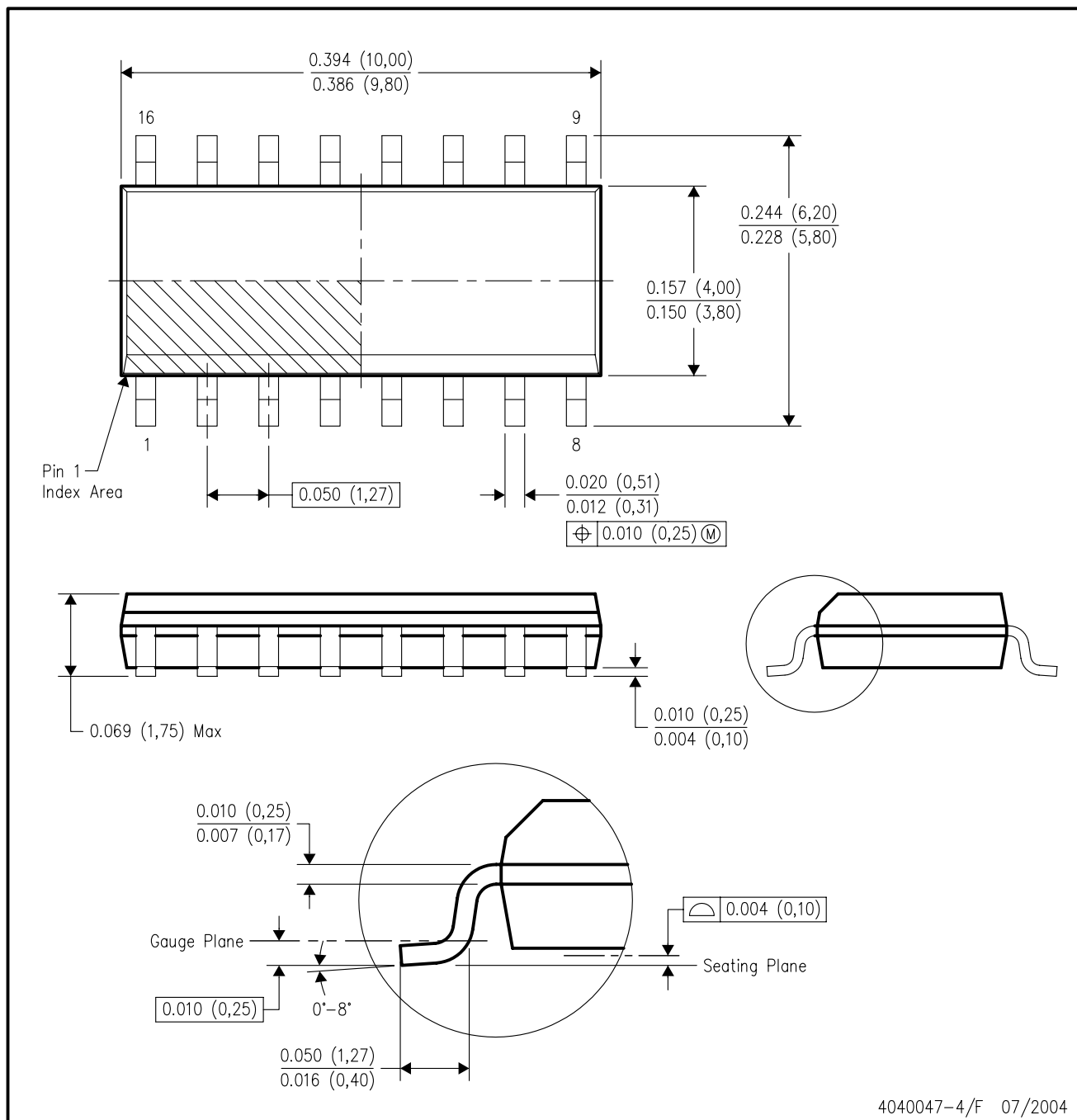


4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 -  Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 -  The 20 pin end lead shoulder width is a vendor option, either half or full width.

D (R-PDSO-G16)

PLASTIC SMALL-OUTLINE PACKAGE



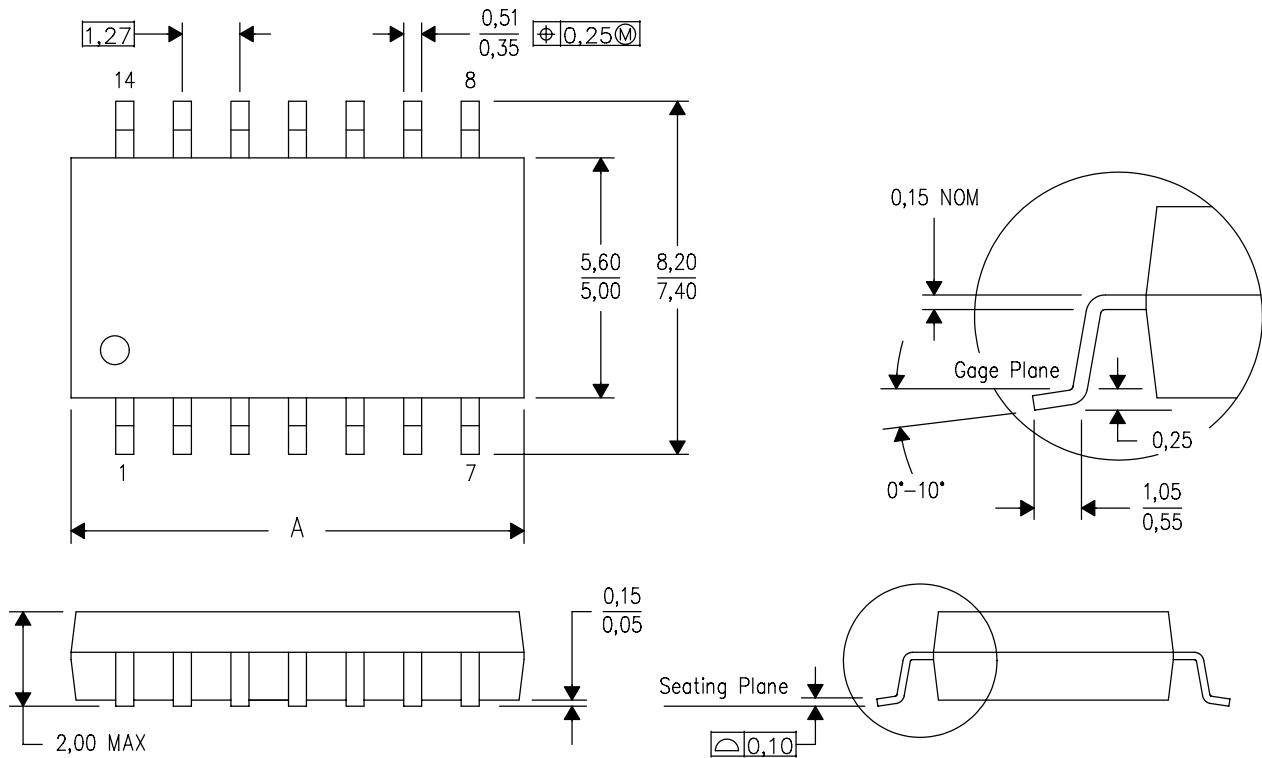
- NOTES:
- All linear dimensions are in inches (millimeters).
 - This drawing is subject to change without notice.
 - Body dimensions do not include mold flash or protrusion not to exceed 0.006 (0,15).
 - Falls within JEDEC MS-012 variation AC.

MECHANICAL DATA

NS (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

14-PINS SHOWN



DIM \ PINS **	14	16	20	24
A MAX	10,50	10,50	12,90	15,30
A MIN	9,90	9,90	12,30	14,70

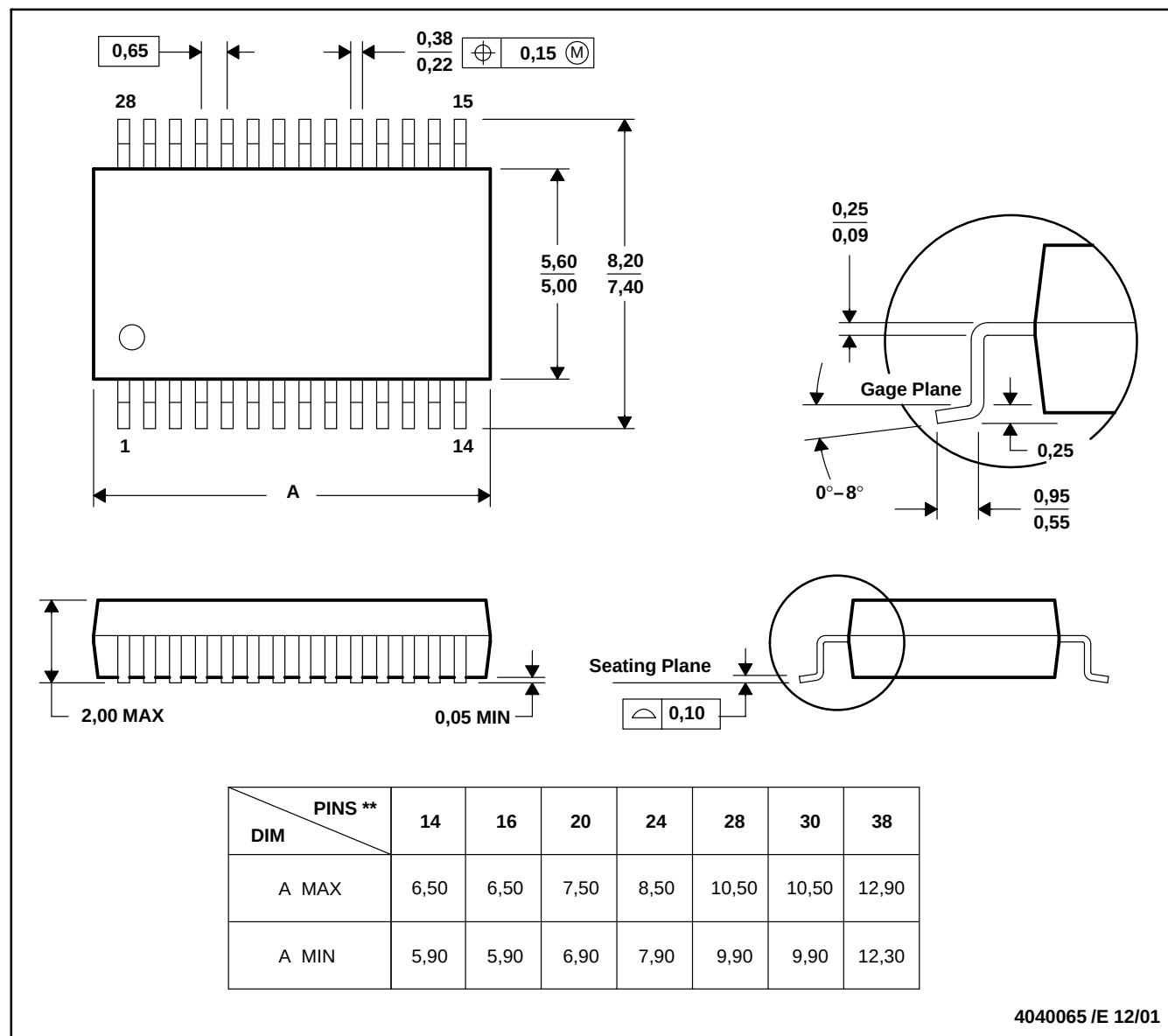
4040062/C 03/03

- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

DB (R-PDSO-G**)

PLASTIC SMALL-OUTLINE

28 PINS SHOWN

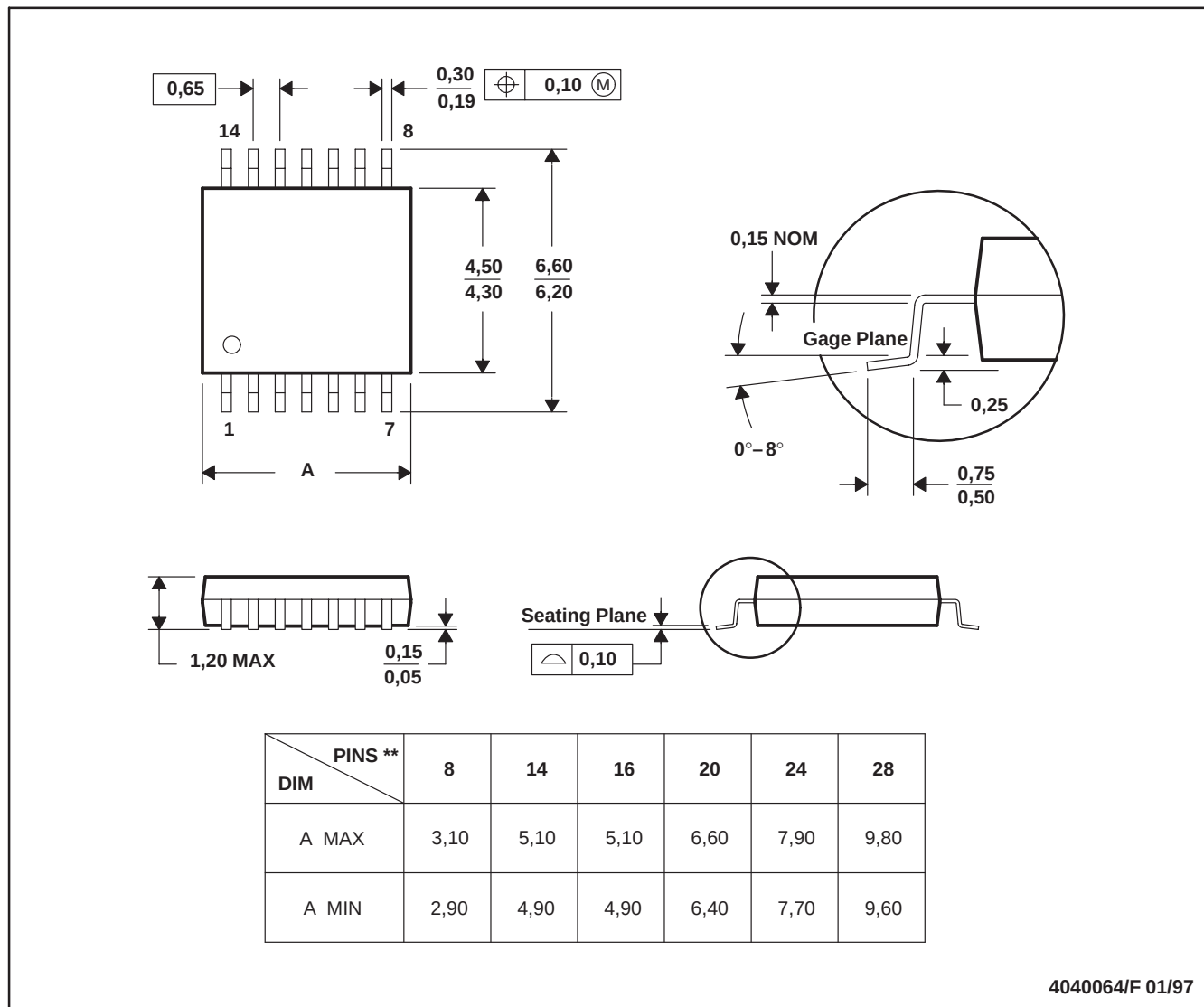


- NOTES: A. All linear dimensions are in millimeters.
 B. This drawing is subject to change without notice.
 C. Body dimensions do not include mold flash or protrusion not to exceed 0,15.
 D. Falls within JEDEC MO-150

PW (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

14 PINS SHOWN



- NOTES: A. All linear dimensions are in millimeters.
 B. This drawing is subject to change without notice.
 C. Body dimensions do not include mold flash or protrusion not to exceed 0,15.
 D. Falls within JEDEC MO-153

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Mailing Address: Texas Instruments
Post Office Box 655303 Dallas, Texas 75265

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